

Bernoulli-EKF scene-graph tracker: per-frame pipeline

1 Scope

Implementation map for one frame of `TwoTierOrchestrator.step()` in `pose_update/orchestrator.py`, with `B-eq:*` cross-references into `bernoulli_ekf.tex` (the formulation paper, authoritative on ambiguity). Per-track state lives in *base frame*: $X^{(i)} = (\ell, T_{bo}, P_{bo}, r, \tau, \mathcal{H})$. SLAM covariance Σ_{wb} never enters the recursion (it appears only in the world-frame emit).

2 Pipeline overview

Standard MOT order, predict-then-measure:

$$\begin{aligned}
 & (T_{wb,k}, T_{bc}(t_k), T_{bg,k}, \text{phase}_k, \{m_l, \hat{\ell}_l, \hat{s}_l, \tau_l\}_l) \\
 & \xrightarrow{\text{0/1 ingest+predict}} (T, P)_{k|k-1} \xrightarrow{\text{2 centroid (cleaned mask)}} \{c_l^{co}\} \\
 & \xrightarrow{\text{3 Hungarian on } d_t^2} \pi^* \xrightarrow{\text{3b ICP at filter prior}} \{(\hat{T}_l^{co}, \hat{R}_l^{co})\}_{l \in \pi^*} \\
 & \xrightarrow{\text{4-6 vis+update+miss}} X_k \xrightarrow{\text{7 pending+admit}} X_k \cup X^{\text{new}} \xrightarrow{\text{8 prune+merge}} \hat{\mathcal{X}}_k
 \end{aligned}$$

Two non-trivial design constraints.

- **Tracker oid is state-only.** The detection’s id (perception’s per-mask tracking id, e.g. SAM2’s `tau`) is metadata, never tracker identity. Tracker oids are minted by `next_track_id()` (fresh consecutive int) at admission only.
- **ICP is a measurement, not an admission gate.** For matched pairs ICP refines (T, R) ; for births ICP runs *exactly once on admission* (centroid-anchored, populates `PoseEstimator.refs[oid]`). The pre-2026 “ICP probes every unmatched detection then admit” path leaked stale reference clouds and is removed.

Per-frame algorithm (one line each).

0. Ingest $T_{wb,k}$, install $T_{bc}(t_k)$ (§3.1); cache $T_{bg,k}$ for held tracks.
1. Predict: static tracks via $u_k = T_{wb,k}^{-1} T_{wb,k-1}$; held tracks via ΔT_{bg} with phase-dependent Q (§4); $r \leftarrow p_s r$.
2. Per-mask cleaned-cloud centroid $c_l^{co} \in \mathbb{R}^3$.
3. Hungarian on the 3-DOF d_t^2 , $\chi_3^2(0.9997) = 21.108$ gate.
- 3b. One ICP per matched pair, seeded with $\hat{T}_{k|k-1,i}^{co} = T_{cb}(t_k) T_{bo,k|k-1}^{(i)}$ against `refs[oid]`.
4. Visibility $p_v^{(i)}$.

5. Matched update: IEKF + Joseph + Huber on d_t^2 ; r via `B-eq:r_assoc`; chain append.
6. Missed: r via `B-eq:r_miss`; T, P unchanged.
7. **Birth via pending buffer:** policy gate (border + tracker-side $n_{\text{obs}} \geq K + \text{score}$) $\rightarrow$ if admitted, `next_track_id()` + ICP once + commit (§3.9).
8. Prune $r < r_{\min}$; greedy self-merge with held-id-as-keeper (§4).
9. Emit $\{i : r \geq r_{\text{conf}}\}$; world-frame composition $T_{wo} = T_{wb}T_{bo}$.

3 Per-frame steps

3.1 Step 0. Ingest + per-frame camera extrinsic

`ingest_slam` caches $T_{wb,k}, T_{wb,k-1}$ for the predict's body-form control $u_k = T_{wb,k}^{-1}T_{wb,k-1}$. The world-form alternative $T_{wb,k-1}T_{wb,k}^{-1}$ produces ~ 1 m drift on real trajectories (adjoint-conjugation mismatch).

The base-to-camera-optical extrinsic $T_{bc}(t_k)$ is *time-varying* on Fetch (head pan/tilt + torso lift). The driver loads it from `pose_txt/T_bc.txt` (one row per frame) and installs it via `set_camera_extrinsic` *before* any measurement-side call. Treating T_{bc} as constant aliases head motion as object motion and silently inflates d_t^2 on every panned frame. Static-predict u_k is T_{bc} -free; only the measurement lift and the ICP prior depend on it.

3.2 Step 1. Predict (state + existence)

Static tracks (the default, `predict_static` with `skip_oids=held_oids`): mean $\mu \leftarrow u_k\mu$, $P \leftarrow \Phi(\text{Ad}(u_k)P\text{Ad}(u_k)^\top + Q)$. `cfg.P_min_diag` optionally floors P 's diagonal per-axis (default `None`; set to ~ 5 mm/0.05 rad to prevent the χ^2_3 gate from spuriously rejecting 2–3 cm same-object drift after $N \gg 1$ updates have shrunk P below per-frame perception jitter). `cfg.P_max` is also `None` by default — saturation Φ becomes the identity unless a cap is configured.

Held tracks (separate branch, `rigid_attachment_predict`): $\mu \leftarrow \Delta T_{bg}\mu$ with $\Delta T_{bg} = T_{bg,k}T_{bg,k-1}^{-1}$, $P \leftarrow \text{Ad}(\Delta T_{bg})P\text{Ad}(\Delta T_{bg})^\top + Q_{\text{manip}}$. Held tracks must be *skipped* from `predict_static`; otherwise the static u_k composes onto $\Delta T_{bg}\mu$ and the kinematics $T_{bo} = T_{bg}T_{go}$ break.

Q is selected by `process_noise_for_phase(phase, is_target, frame, frames_since_obs)`:

- *Static* ($\neg \text{is_target}$): Q decays with `frames_since_obs` from $\mathcal{N}(0, 10^{-6})$ down to $\mathcal{N}(0, 10^{-8})$ per axis.
- *Held, holding*: $Q = \text{diag}(2.5 \cdot 10^{-4}, 10^{-3})$ per trans / rot axis. Models grip slack + proprioception jitter.
- *Held, grasping/releasing*: $Q = I$ (essentially unknown). The transition frame's pose is not a function of ΔT_{bg} .

Bernoulli predict: $r_{k|k-1} = p_s r_{k-1|k-1}$ (default $p_s = 1$).

3.3 Step 2. Coarse measurement (cleaned mask centroid)

For each detection: $c_l^{co} = \text{centroid_cam_from_mask}(m_l, D_k, K)$. The mask is first passed through `clean_mask` (`pose.update/icp_pose.py`):

1. Binary erosion (1 pixel) drops the worst boundary band.

2. Per-masked-pixel 4-neighbour depth-jump rejection ($|\Delta d| > 3$ cm kills boundary pixels straddling foreground/background; object curvature is sub-cm).

Both stages *self-disable* below 30 surviving pixels (small masks keep their thin rim). On `apple_in_the_tray` frame 300 (a ~ 15 k-pixel tray detection): retains 90% of pixels, shrinks `obj_radius` from 19.9 to 16.4 cm by removing background leakage tails. Centroid shift is sub-mm (filter is spatially symmetric); ICP fitness stable.

The Hungarian cost uses isotropic $R_{\text{cent}} = \text{diag}((2 \text{ cm})^2 I_3)$. Detections that fail back-projection (< 30 valid-depth pixels even after cleaning) are dropped from association *and* from the birth buffer.

3.4 Step 3. Hungarian association

`association.hungarian_associate`, called with a 3-DOF centroid innovation:

$$\nu = [\nu_{3D}, \mathbf{0}]^\top \in \mathbb{R}^6, \quad S = \text{diag}(S_{3D}, \tau I_3), \quad (1)$$

$$d_t^2 = \nu_{3D}^\top S_{3D}^{-1} \nu_{3D}, \quad d_r^2 = 0 \quad (2)$$

with $\tau = 10^6$ in the rotation slot to make the block decomposition cleanly 3-DOF. $S_{3D} = P_{tt}^{(i)} + R_{\text{cent}}^{bo}$ where $R_{\text{cent}}^{bo} = R_{bc} R_{\text{cent}}^{cam} R_{bc}^\top$ is the rotation-only base-frame lift of the camera centroid covariance. A naive 6-DOF $R_{icp} = \text{diag}(R_t, \tau I)$ would couple τ into translation via $[t_{bc}]_x$ and collapse $d_t^2 \rightarrow 0$.

Cost (in χ^2 units, `cost_d2_mode='sum'`):

$$C_{il} = (d_{il,t}^2 + 0.1 d_{il,r}^2) - \alpha \mathbb{1}[\tau_l = \tau^{(i)}, \tau^{(i)} \geq 0] + \beta \mathbb{1}[\hat{\ell}_l \notin \mathcal{H}^{(i)}] + \gamma \max(0, 1 - \hat{s}_l) \quad (3)$$

with $\alpha = 4.4$ (SAM2-id continuity, fixed). β and γ are `BernoulliConfig` knobs (*defaults* 0.0 — soft path disabled): `hungarian_label_penalty` and `hungarian_score_weight`. `tests/visualize_ekf_tracking.py` overrides them to $\beta = 6.0$ (label-flicker soft penalty, $< G_{\text{out},t}$ so geometry can override label) and $\gamma = 2.0$ (low-score mild penalty). Soft mode requires `enforce_label_match=False`; the default `enforce_label_match=True` forces label-mismatch cells to `_INFEASIBLE` = 10^{12} regardless of β .

Two outer gates (in addition to the soft cost):

1. Probabilistic chi²: $d_{il,t}^2 \leq G_{\text{out},t} = 21.108$ (χ_3^2 at 99.97%, `gate_mode='trans'`).
2. Hard Euclidean residual cap: $\|\nu_{3D}\|_2 \leq \text{cfg.max_residual_m} = 0.30 \text{ m}$. Independent of the chi² gate. Catches the inflated-covariance pathology where, after a long miss run with $\sigma_t \approx 0.4 \text{ m}$, an 0.86 m residual gives $d_t^2 \approx 17 < 21.108$ and the chi² gate becomes meaningless. Set to `None` to disable.

Cells failing either gate get the sentinel `_INFEASIBLE`.

Label-purity gate (soft mode only). The membership test $\hat{\ell}_l \notin \mathcal{H}^{(i)}$ uses `_label_in_history_meaningful(t_histod_label, min_obs=5, min_share=0.05)` from `pose_update/association.py`: a label counts as “in the history” only if it has ≥ 5 observations *and* $\geq 5\%$ share of the total. This rejects stale trace-amount labels (e.g. 3 obs against 53 990) that would otherwise bypass the β penalty.

Solver: `scipy.optimize.linear_sum_assignment` on the rectangular cost matrix; assignments at `_INFEASIBLE` are post-filtered out (Hungarian on a fully-infeasible row will still emit some column). Result: `match{oid: det_idx}`.

Recovering the paper baseline: `enforce_label_match=True`, $\beta = \gamma = 0$ (the default), `gate_mode='full'`, `cost_d2_mode='full'`. **Oracle mode:** `association_mode="oracle"` bypasses the cost matrix and matches by upstream `det["id"]`; used by the degeneracy test.

3.5 Step 3b. Per-match fine ICP

For every $(i, l) \in \pi^*$: `PoseEstimator.estimate(oid=i, mask=m.l, depth=D, T_co_init=  $\hat{T}_{k|k-1,i}^{co}$ )`. The seed is the matched track’s filter prior; ICP refines against `_refs[i]` (*tracker-oid-keyed*, stable across SAM2 id reissues for the same physical object).

Output $\hat{R}^{co}$ uses data-driven floors:

$$\begin{aligned}\sigma_t^2 &= \max(\text{rmse}^2, (2\text{ cm})^2) \cdot \text{fit_scale}, \\ \sigma_r^2 &= \max((\text{rmse}/r_{\text{obj}})^2, (0.1\text{ rad})^2) \cdot \text{fit_scale}.\end{aligned}$$

ICP failure ($\text{fitness} < 0.90$ or $\text{rmse} > 15\text{ mm}$) demotes the pair to a centroid-only update with inflated rotation R (no rotation information through Kalman).

3.6 Step 4. Visibility $p_v^{(i)}$

$p_v^{(i)} = \mathbb{1}[\pi_K(T_{bc}^{-1}T) \in \Omega] \cdot \prod_{j \neq i} (1 - \rho_{ij})$ (§). Implementation: `visibility.visibility_p_v`. Off when `enable_visibility=False`; Production orchestrator still uses constant T_{bc} here (test driver uses per-frame).

3.7 Step 5. Matched-track update

For matched (i, l) : lift the camera-frame measurement to base via $T_{bo,\text{meas}} = T_{bc}(t_k)\hat{T}^{co}$, $R^{bo} = \text{Ad}(T_{bc})\hat{R}^{co}\text{Ad}(T_{bc})^\top$. Then IEKF (2 inner iterations) + Joseph form + Φ saturate + diagonal floor on P .

Existence update: $r \leftarrow \text{r_assoc_update_loglik}(r, \log \mathcal{L}, p_d, \lambda_c)$ in log-space (avoids underflow for $\log \mathcal{L} \sim -50$).

Robust gate: Huber weight on d_t^2 ($G_{\text{in},t}$ inner soft, $G_{\text{out},t}$ outer hard); $w = 0$ tracks revert to the miss branch. Append $(k, \hat{T}^{co}, \hat{R}^{co})$ to the loop-closure-aware chain $\mathcal{H}^{(i)}$ (§).

3.8 Step 6. Missed-track update

For unmatched i : $r \leftarrow (1 - \tilde{p}_d)r / ((1 - \tilde{p}_d)r + (1 - r))$, $\tilde{p}_d = p_d p_v^{(i)}$. T, P unchanged. When $p_v \rightarrow 0$ the update collapses to $r_{k|k} = r_{k|k-1}$.

3.9 Step 7. Birth via pending buffer

Two-phase, with the *tracker* owning admission counters:

Pending buffer (`self._pending_births`, dict keyed by perception id). For each unmatched detection l that passes the back-projection gate:

1. Look up / create `PendingBirth(source_id=det["id"], ...)`; bump `n_obs_tracker`, update `max_score`.
2. **Policy gate** (`birth_admissible` with `require_pose=False`): `bbox` not on image border ($\geq 2\text{px}$), `n_obs_tracker` $\geq K = 3$, `max_score` $\geq S = 0.20$. Note: K counts the *tracker*’s unmatched-streak frames, not perception’s `n_obs` (which resets on SAM2 id reissue).
3. If admitted: mint $i_{\text{new}} = \text{next_track_id}()$ (fresh consecutive int, NOT seeded from `det["id"]`); call `PoseEstimator.estimate(oid=i_new, T_co_init=None)` once (first-observation branch: anchors at c_l^{co} , populates `_refs[i_new]`); commit via `_birth_track`; pop the pending entry. Bernoulli init: $r = \lambda_b \hat{s} / (\lambda_b \hat{s} + \lambda_c)$.

4. If rejected: keep the pending entry. TTL-expire after `birth_pending_ttl_frames=30` of no re-arrival (one second at 30 Hz; survives short perception dropouts).

This invariant matters: $i \in \text{refs}$ iff i is a committed track. The pre-2026 path called `estimate` as an admission probe, seeding `refs[i]` before the policy gate and leaking it on rejection; subsequent frames then ran refine-ICP against that stale cloud and silently dropped real birth candidates (manifested as the apple-in-tray + tray failing to ever birth).

Birth covariance: by default (`init_cov_from_R = True`, the `BernoulliConfig` default) $P_{\text{new}} = \hat{R}^{co}$ — the ICP-derived measurement covariance from the on-admission `PoseEstimator.estimate` call. Setting `init_cov_from_R = False` falls back to the fixed $\text{diag}((5\text{ cm})^2, (0.05\text{ rad})^2)$.

3.10 Step 8. Prune + self-merge

Prune (`_prune_track`): drop i with $r < r_{\min} = 10^{-3}$ from `state.objects`, all per-orchestrator dicts, and `refs[i]`.

Self-merge (`_self_merge_pass(held_id)`): Hungarian’s one-to-one constraint leaves duplicates when SAM2 emits parallel ids for one physical object. Per-frame greedy fusion of same-label pairs with $\|\mu_i - \mu_j\| \leq \text{self_merge_trans_m} = 5\text{ cm}$. Keep decision: held oid wins (preserves rigid-attachment kinematics); otherwise longer chain wins; tie-break lower oid. Fusion is Bayesian information sum on $\text{SE}(3)$ linearised at μ_{keep} :

$$P_{\text{new}}^{-1} = P_{\text{keep}}^{-1} + P_{\text{drop}}^{-1}, \quad \mu_{\text{new}} = \mu_{\text{keep}} \text{Exp}(P_{\text{new}} P_{\text{drop}}^{-1} \text{Log}(\mu_{\text{keep}}^{-1} \mu_{\text{drop}})). \quad (4)$$

Bookkeeping: $r \leftarrow \max$, `frames_since_obs` $\leftarrow \min$, `label_scores` per-label sum + sample-weighted mean, $\mathcal{H}^{(\text{drop})}$ appended to $\mathcal{H}^{(\text{keep})}$, `refs[drop]` merged into `refs[keep]` via voxel-downsample.

3.11 Step 9. Emit

`TwoTierOrchestrator.objects` returns $\{i : r^{(i)} \geq r_{\text{conf}} = 0.5\}$. Tentative tracks are still predicted/associated/updated but hidden; visible via `tentative_objects`. World-frame composition $T_{wo} = T_{wb} T_{bo}$ (or the chain smoother of § when bundle-adjusted).

4 Held-object handling

The dataset has no annotated phase; the test driver’s `_GripperStateInferer` (`tests/visualize_ekf_tracking.py`) infers `{phase, held_obj_id}` from proprioception:

Phase from gripper finger width. $w = \text{l_gripper_finger_joint} + \text{r_gripper_finger_joint}$ (from `joints_pose.json`). Hysteresis bands prevent chatter:

$$\text{is_closed}_k = \begin{cases} w_k < 0.040\text{ m}, & \text{previously closed,} \\ w_k < 0.025\text{ m}, & \text{previously open.} \end{cases}$$

Project assumption. The gripper begins fully open; ANY closing of the fingers registers as a grasp action. The FSM additionally fires whenever w_k drops more than `close_delta_m=0.005 m` below the rolling open baseline, regardless of the absolute thresholds — a hard-coded 0.025 m boundary would miss objects that the gripper clamps onto without fully closing. Phase transitions: `{open  $\rightarrow$  closed} = grasping`, `{closed  $\rightarrow$  closed} = holding`, `{closed  $\rightarrow$  open} = releasing`, otherwise *idle*.

Held identity from EE proximity. At grasp onset (idle→grasping) compute the world-frame EE position $T_{we} = T_{wb}T_{bg}$ and pick the nearest live track within `grasp_radius_m=30` cm; that becomes `held_obj_id`. The id is *frozen* through holding (no per-frame re-inference – that would jitter as masks resegment), released at the *releasing* edge, and migrated through self-merges via `apply_merges` (which reassigns to the keeper if the held oid was a drop).

Q schedule. The orchestrator passes $\{T_{bg}, \text{held_oids}, \text{phase}\}$ to `tracker.step(). rigid_attachment_predict` reads the phase-dependent Q from `process_noise_for_phase(phase, is_target=True, frame="base")`: huge during grasping/releasing (transition pose is unknown), ~ 1.6 cm/ 1.8° per frame during holding (grip slack + proprio jitter). The pre-fix $\mathcal{N}(0, 10^{-8})$ value was unrealistically tight; P stayed pinned at the floor and the Hungarian gate rejected every re-detection of the moving held object.

Self-merge protection. `_self_merge_pass(held_id=...)` forces the held oid as keep in any pair it participates in, preserving the rigid-attachment motion model across merges.

5 Code map (quick reference)

Step	Equation / role	Code pointer
0	$T_{bc}(t_k)$ install	<code>GaussianState.set_camera_extrinsic</code>
1	Exp / Log, predict, Φ	<code>ekf_se3.{se3_exp, se3_log, saturate_covariance}</code>
1	static + held predict	<code>predict_static, rigid_attachment_predict</code>
1	Q schedule	<code>ekf_se3.process_noise_for_phase</code>
1	$r_{k k-1} = p_s r$ (B-eq:bern_pred_r)	<code>bernoulli.r.predict</code>
2	cleaned mask + centroid	<code>icp_pose.{_clean_mask, centroid_cam_from_mask}</code>
3	cost + Hungarian (B-eq:cost_sam2, B-eq:hung)	<code>association.hungarian_associate</code>
3	label-purity gate	<code>association._label_in_history_meaningful</code>
3b	per-match ICP	<code>PoseEstimator.estimate</code> (refine branch)
4	p_v (B-eq:pv)	<code>visibility.visibility_p_v</code>
5	innovation, IEKF update, Joseph, Huber	<code>GaussianState.update_observation, ekf_se3.huber_weight</code>
5	r (B-eq:r_assoc)	<code>bernoulli.r_assoc.update_loglik</code>
6	r (B-eq:r_miss)	<code>bernoulli.r_miss.update</code>
7	policy gate	<code>birth_admissible</code> (require_pose=False)
7	pending buffer	<code>_PendingBirth, _pending_births</code> dict
7	admit + commit (B-eq:birth_r, B-eq:birth_state)	<code>_next_track_id + _birth_track</code>
7	ICP at admit (first-obs branch)	<code>PoseEstimator.estimate</code> (T_co_init=None)
8	prune	<code>_prune_track</code>
8	self-merge (B-eq:self_merge_info)	<code>_self_merge_pass(held_id=...)</code>
9	emit $r \geq r_{\text{conf}}$	<code>TwoTierOrchestrator.objects</code>
held	phase + held oid inference	<code>_GripperStateInferer</code>

6 Known deviations from `bernoulli_ekf.tex`

1. Standalone `ekf_se3.ekf_update` uses $(I - K)P$ rather than Joseph and skips Φ . The RBPF Bernoulli path (`RBPFState.update_observation`) is paper-correct.
2. Production orchestrator still passes constant T_{bc} to the visibility predicate; the test driver installs per-frame $T_{bc}(t_k)$ via `set_camera_extrinsic`. Migrate by adding the call in `step()` before any measurement-side function; until then, set `enable_visibility=False` on the production path or accept head-pan-aliased visibility flicker.

3. Visibility footprint $A^{(i)}$ uses the matched-detection bbox as a proxy. Paper prescribes projecting the TSDF shape summary $\mathcal{G}^{(i)}$.
4. *Mask-edge cloud filter* (§3.3) and *rigid-attachment predict with phase-dependent Q* (§4) are manipulation-aware extensions not in the paper.
5. *Pending-birth buffer* (§3.9) replaces the paper’s per-frame “unmatched detection $\rightarrow$ immediate birth” with a tracker-side admission policy. The Bernoulli existence r semantics are unchanged; only the gating before r_{new} is computed differs.
6. *Hard Euclidean residual cap* (`BernoulliConfig.max_residual_m=0.30 m`, §3.4): on top of the χ^2 gate, an absolute Euclidean residual cap on $\|\nu_{3D}\|_2$ rejects matches that pass the probabilistic gate only because the prior covariance has inflated. Independent of any β/γ soft-mode setting.
7. *Soft label-purity gate* (`_label_in_history_meaningful`, §3.4): when `enforce_label_match=False`, the soft β -penalty is bypassed only for labels with ≥ 5 observations *and* $\geq 5\%$ share of the track’s history — pure dict membership would be triggered by 3 stale obs against tens of thousands of correct ones, defeating the soft penalty.
8. *Gravity-aware predict at release* (§): the orchestrator detects the FSM exiting {holding, releasing} and replaces the just-released oid’s (T_{wo} , P_{wo}) with the post-fall prediction from `pose_update/gravity_predict.py`. Voxel observability (`pose_update/voxel_observability.py`) supplies the supporting surface; `pose_update/object` resolves per-label dynamics (e , μ , shape, r_{obj}). The pre-2026 release-time Q was a generic “essentially unknown” inflate that left the EKF mean at the gripper height, so the next re-observation had to apply a 30 cm correction in one step. With the gravity hook the mean lands at the predicted resting-pose, inflated only by the parametric (σ_{xy} , σ_z , σ_{yaw}). Gated by `BernoulliConfig.gravity_predict` and `TwoTierOrchestrator.voxel_obs` (None disables the hook).